Chandana Galgali

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EDUCATION

K J Somaiya College of Engineering, Somaiya Vidyavihar University, Mumbai, Maharashtra, India
Bachelor of Technology (Information Technology), Honors in Artificial Intelligence | 2022–2026
Overall CGPA: 9.67/10 • Honors CGPA: 9.40/10 • Department Rank 1

- Engineered an interactive virtual laboratory to simulate core DSA, enhancing learning for 350+ second-year students.
- Secured ₹1.5 Lakh grant to deploy an AI-powered hyperlocal air quality monitoring platform across the campus.
- Led a 30-member council, organizing technical events that drove 500+ footfall and secured ₹20,000+ in sponsorships.

WORK EXPERIENCE

Indian Institute of Technology Bombay, Mumbai, Maharashtra, India
Compiler Optimization Research Intern | Jan 2026 - Apr 2026

- Developed a scope-aware static analysis framework for JavaScript JIT compiler optimizations, focusing on AST optimizations.
- Integrated an approximate interpretation pipeline to generate Ahead-of-Time behavioral hints, demonstrating upto 45% execution speedup and 48% reduction in peak memory usage.
- Instrumented QuickJS engine internals with custom runtime hooks to validate static hints against runtime ground truth, attaining 97.5% property mutation recall across micro-benchmarks.

Barclays Global Service Centre Private Limited, Pune, Maharashtra, India
Software Development Engineer Intern | Jun 2025 - Jul 2025

- Built a real-time SLA monitoring tool for regulatory feed reporting, reducing manual validation time from 45 minutes to under 2 minutes. Developed a predictive PoC using unsupervised learning on incident logs to flag high-risk system failures.
- Developed an automated Ansible housekeeping system to resolve recurring pod failures and NAS storage issues.
- Served as Scrum Master, facilitated agile ceremonies, and presented technical solutions to CIOs and Managing Directors for high-level leadership visibility.

Keva - Fragrances, Flavours & Aroma Ingredients, Mumbai, Maharashtra, India
Web Application Security Intern | Jan 2025 - Mar 2025

- Executed security testing and code quality audits using SonarQube and Burp Suite on 2 web applications across frontend and backend to mitigate SQLi, CORS and CSRF risks.
- Discovered a high-severity admin access flaw and a CWE-319 cleartext transmission risk via manual vulnerability testing and compiled comprehensive executive audit reports.

Java Developer Intern | Jan 2023 - Mar 2023

- Engineered HashMap-based Java backend algorithms to map chemical replacement logic, optimizing manual formula checks down to O(1) runtime execution.

PROJECTS

AI-Powered Multi-Pollutant Mapping for Hyperlocal Air Quality Monitoring (1st Place, Climate Hackathon 2024)
Python, Flask, Convolutional Neural Networks (CNN), Google Earth Engine, TensorFlow, React

- Utilized Sentinel-5P and Sentinel-2 satellite datasets to downscale coarse-resolution pollutant concentration data via statistical correlation, creating hyperlocal pollutant maps with an improved spatial resolution from 1000m to 10m with an 88.88% accuracy rate for urban pollution hot-spot identification.

Text to AR/VR: Generating AR/VR Visualization from Textual Description of Numerical Data
Python, Django, Large Language Model (LLM), Retrieval-Augmented Generation (RAG) Model, Unity

- Developed a novel Text-to-AR/VR pipeline using LLMs and a custom RAG framework over a self-curated dataset, transforming textual descriptions of numerical data into interactive and immersive 3D visualizations via direct natural language parsing to Unity spatial rendering with an 88.89% accuracy rate.

Voice-Enabled Geospatial UI (ISRO Bharatiya Antariksh Hackathon 2024)
JavaScript, Transformers.js, WebGL, Leaflet.js, Hugging Face, OpenStreetMap

- Engineered an offline Leaflet.js GIS web app utilizing a local Whisper model (~73MB) via WebGL for <600ms speech-to-text latency. Integrated an intent-driven NLP engine (94.5% accuracy) with the Overpass API mapping 25+ OpenStreetMap tags, enabling hands-free spatial data discovery and client-side PDF reporting.

Automated Caption Generation using Encoder-Decoder Model for Targetted Objects (Jewellery Images)
Python, FastAPI, VGG-16, GRU, PyTorch, Torchvision, React
- Engineered a deep learning system (VGG-16 & GRU) trained across 32,190 augmented images to generate structured image descriptions of targeted objects in 1-2 seconds, achieving 94.62% detection and 90.44% captioning accuracy.

ADDITIONAL PROJECTS

- **AST Query:** Built a zero-dependency Python Static Analysis CLI to query and filter AST structures.
- **Adaptive UI Theme React Component:** Published a hardware-aware, lux-toggling React componen to npm.
- **Machine Learning & CV Portfolio:** Built predictive and vision models using LSTM, SVM, K-Means and OpenCV.
- **Automation Utilities:** Developed workflow scripts for bulk email purging and batch image optimization.

TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, TypeScript, C++, C, Java, R, SQL
- **AI, Machine Learning & Data Science:** TensorFlow, PyTorch, Keras, Scikit-Learn, Hugging Face, Transformers, OpenCV, CNNs, RNNs (LSTM, GRU), VGG-16, LLMs, RAG, Pandas, NumPy, Predictive Algorithms (K-Means, SVM, Linear Regression)
- **Backend & Web Development:** React.js, Next.js, Angular, Node.js, Flask, FastAPI, Django, Spring Boot, GraphQL, WebGL, HTML5/CSS3
- **DevOps, Security & Cloud:** Docker, Kubernetes, Ansible, Git (GitHub/GitLab CI/CD), SonarQube, Burp Suite, Metasploit, FortiSIEM, Red Hat OpenShift, Linux/Ubuntu, AWS (Botocore)
- **Databases & Data Management:** PostgreSQL, MySQL, MongoDB, SQL Server Management Studio
- **Specialized Frameworks, Tools & Platforms:** Unity (AR/VR), Babel (AST Parsing), Leaflet.js, Google Earth Engine, MATLAB, RStudio, npm, Jira, Confluence, Agile/Scrum, VSCode, Docker Desktop

ACHIEVEMENTS

First Place & ₹1.5 Lakh Grant, Climate Hackathon 2024, Issued by the U.S. Consulate General (Mumbai) and Ekonnect Knowledge Foundation
- Led backend development in a 4-person team, outperforming 150+ competing teams to secure 1st place.
- Awarded a ₹1,50,000 grant to deploy the AI-powered multi-pollutant mapping system for hyperlocal air quality monitoring across the 65-acre campus (39,000+ students) under the “Towards Green Campuses” initiative.

CERTIFICATIONS

AI, Cloud & Data Science:
- Microsoft: Introduction to Generative AI for Data Analysis | Introduction to Microsoft Azure Cloud Services
- University of Colorado Boulder: Introduction to Computer Vision | Modeling of Autonomous Systems
- The State University of New York: Big Data, Genes, and Medicine

Software Engineering & Application Development:
- Duke University: Java Programming: Solving Problems with Software | Java Programming: Arrays, Lists, and Structured Data
- Meta: Programming Fundamentals in Kotlin | Introduction to Android Mobile Application Development
- Vidyalankar Polytechnic: Computer Fundamentals

Cybersecurity, Hardware & DevOps:
- IBM: Introduction to DevOps
- Google: Foundations of Cybersecurity | Play It Safe: Manage Security Risks
- University of Maryland: Hardware Security

Blockchain & Web3:
- University at Buffalo: Blockchain Basics | Smart Contracts | Decentralized Applications (Dapps)
- LearnQuest: Blockchain Security Advanced Protection

Finance & Business:
- University of Geneva: Understanding Financial Markets
- University of Virginia: Financial Accounting Fundamentals | Digital Product Management: Modern Fundamentals

Languages & Soft Skills:
- UC Davis: Learn Spanish: Basic Spanish Vocabulary